Twenty-four minutes of warning: what a reported lead time licenses, and what reporting standards do not ask

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Abstract

A median alert lead time is a capability claim: a system reported to warn a median of 24 minutes before an event invites the reading that it will usually give about that much warning. That is not what the number measures on its own. A first threshold crossing can only occur inside the pre-onset window over which a patient was observed. The statistic is therefore set jointly by the model and by the cohort's distribution of those windows, and a single median cannot separate the two. That makes it a reporting problem before it is a modeling one. That a lead time cannot exceed the window it is searched in is definitional; what has not been measured is how strongly the reported first-crossing lead moves with realized, patient-varying available-window length in realistic evaluation pipelines. Across three perioperative cohorts and five model families, at thresholds transported unchanged from a development validation split, median lead among detected event cases rises from the shortest to the longest realized pre-onset window band in all fifteen model-by-cohort combinations. The rise is 26 to 54 minutes. Read beside its operating point, the longest pooled median lead here, 24 minutes — a median over the event cases a model detected — stands beside a case-level specificity of 0.406 measured over all control cases; the minimum specificity across the fifteen combinations is 0.379. Two natural responses leave the dependence in place; a horizon-specific detection rate reduces it without removing it, and restricting to a constant window removes it but reveals that the longest median co-occurs with the lowest detection rate. In our author-completed TRIPOD+AI item-level review, the closest requests are items 20a and 20b, which ask for follow-up time as a cohort descriptor; we did not identify an item that asks for the distribution of the window the alert rule actually searches, for a window-conditional timing measure, or for a lead-time summary to be interpreted jointly with its realized operating-point metrics. We set out four candidate disclosures for continuously monitored models, task-specific complements to TRIPOD+AI. A median lead remains a descriptive interval among detected cases, but on its own it supports neither a model-intrinsic nor a cohort-comparable reading of warning capability. We analyze what a reporting convention makes inferable, not what individual readers infer, and do not measure comprehension here.

1 Introduction

A number that travels. A median alert lead time computed on a held-out split need not stay in the paper that produced it. It can travel into a summary written for a purchaser, a press description, and finally the sentence a clinician uses to say what the monitor on the wall does. Qualifiers are easy to

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shed at each hop, and the label survives them because it is already in plain language: the name of the metric asserts, with no further explanation, that the system provides that much warning. Reporting standards decide what a reader is guaranteed to receive alongside such a claim. Throughout, what a number **licenses** is what a reader is entitled to conclude from it plus the disclosures a standard guarantees, and an **affordance** of a label is a reading it thereby makes available: both are properties of the convention, not measurements of what readers conclude (Section 7). This paper asks what that guarantee currently is for one such metric under one general standard, and finds it insufficient to support the reading the label invites.

Why a warning interval matters here. Mean arterial pressure is a routinely monitored hemodynamic measure during surgery. Sustained falls below roughly 65 mmHg are intraoperative hypotension, and accumulated exposure is associated with kidney and myocardial injury after surgery [Walsh et al., 2013, Salmasi et al., 2017, Wesselink et al., 2018, Sessler et al., 2019]. The responses are ordinary anesthetic care, and none acts the instant it is chosen, so a warning interval is worth what a clinician can choose and deliver inside it. That is why monitors here are evaluated on how far ahead they alert [Hatib et al., 2018, Davies et al., 2020, Wijnberge et al., 2020, Maheshwari et al., 2020]. This paper is about that reported interval, not about physiology (Section 4.2).

A model that is read continuously is asked how early it warns. The **median alert lead time** is the statistic reported in answer across acute care: the interval between the first minute a case's score crosses an alert threshold and the onset of the event. We have not surveyed how often it is reported, so what follows is a multi-cohort case study rather than a field-level audit. It appears for sepsis [Wong et al., 2021], for acute kidney injury [Lee et al., 2026], and for intraoperative hypotension, where a commercial index is evaluated on the interval between its alert and the event. The statement that a system has a **median alert lead time of 24 minutes** invites the reading that it will usually give around twenty-four minutes of warning.

The definition is the same in every one of those settings, and it carries a dependence its name conceals. The first crossing is searched for inside the interval the patient was observed, so the lead is bounded above by that window. The window's length is set by event timing relative to eligibility, by monitoring practice and by case mix, not by the model. Table A1 sets each reading the label invites beside what the number can carry, and the disclosure that would separate them.

What is already known, and what is new here. Criticism of early-warning evaluation is not new: the C-statistic has been argued uninformative for early warning scores, with per-alert and workload measures proposed instead [Romero-Brufau et al., 2015]; monitor alerting has been quantified as an alarm burden per patient-day [Paine et al., 2016]; choice of evaluation sample moves reported performance for this exact task on these public data [Yang et al., 2025]; and the task's own retrospective traces its arc from hope to hype [Michard and Futier, 2023]. And cross-center evaluation separates learned methods from pressure-derived rules far less cleanly than single-center reports suggest [Chaari et al., 2025]. Closest to this paper, Lauritsen et al. [2021] names the temporal framing choices — observation, prediction and lead windows, window shift, event trigger — and shows discrimination and calibration moving with them, asking that a standard record those choices. Its windows are set by the developer, and what it measures moving is discrimination and calibration. That bound is likewise definitional, and Section 3.3 treats it as such. What is not established is **how strongly a reported first-crossing lead time moves with the realized, patient-varying window it was measured over**, and the substitutes those critiques recommend are what Section 6.2's second item asks for.

What this paper contributes. This is a reporting-standards contribution, evidenced by a multi-cohort measurement study. We map the case study onto an author-completed TRIPOD+AI [Collins et al., 2024] checklist, which did not request the disclosures that would have made any of this visible. Alongside it we propose four candidate additions for continuously monitored models (Section 6). Three measurements stand behind that proposal. We measure the dependence, in the metric's own units and against two explicit references rather than against the ceiling (Section 3). We report the operating point beside the lead, which no lead-time statistic carries, and find the two run opposite across combinations (Section 4). And we implement the two repairs the first two sections invite, neither of which works, while a constant-window restriction removes the dependence and reveals that the longest median co-occurs with the lowest detection rate (Section 5). The argument is about a reported quantity, not about how well any model discriminates, which is not in dispute.

2 Setting

Task and outcome. We define intraoperative hypotension as mean arterial pressure below 65 mmHg sustained for at least three consecutive minutes. The 65 mmHg level is conventional in this literature [Salmasi et al., 2017, Sessler et al., 2019], which characterizes exposure cumulatively rather than as a consecutive run; the three-minute sustain is this study's operationalization. Mean arterial pressure is also one of the models' input channels, so the label is computed from a signal the models read; Sections 4.2 and 7 say what that does not license.

Cohorts. Development used MOVER's EPIC arm; we evaluate on its held-out test split (1,404 cases, 786 with a sustained event) and on two external cohorts, VitalDB [Lee et al., 2022, Lee and Jung, 2022] with 2,804 cases and 1,007 events, and the SIS-derived arm of MOVER [Samad et al., 2023, UCI Machine Learning Repository, 2023] with 2,641 cases and 1,409 events. MOVER's two arms come from one site and were de-identified separately, so patient overlap between the development cohort and MOVER (SIS) cannot be measured and is not excluded; VitalDB is the external cohort with no such question. MOVER (SIS) releases no usable subject key, so cases are the clustering unit there and subjects in the other two; all intervals in this paper are 95% cluster bootstrap on that unit unless a caption says otherwise.

Models. Five families: a transformer over irregularly sampled triplets [Tipirneni and Reddy, 2022], referred to throughout as the ITE (initial triplet embedding) Transformer, an LSTM, XGBoost, random forest and logistic regression. Two read the window as a sequence and three a per-minute feature vector; Appendix A.1 has the architectures and training. No model is proposed for clinical use and none of what follows is a ranking.

Deployment simulation. Every model is read forward at **every eligible minute of every case**, using only measurements recorded strictly before that minute and after a source-specific induction exclusion (Appendix A.1).

Operating points. An **operating point** is the score threshold at which the system is read as alerting, with the sensitivity and specificity that follow. **Case-level sensitivity** is the share of event cases that alert at some point in their alert window and **case-level specificity** the share of control cases that never do. Calibration and thresholds were fitted on the development cohort's validation split at an 80% case-level sensitivity target and then **transported** unchanged to every other cohort. Nothing here reselects a threshold, split, seed or operating point, and no model is retrained or re-scored.

The alert rule and the window it searches. A case alerts if its score reaches the threshold at any minute in the 60 minutes before the event, and its lead time is the interval from the first such minute to the event. The interval the rule searches is the case's **available pre-onset evaluation window**: its realized observable run-up to onset, after the induction exclusion, capped at the 60-minute alert window. A short window is therefore not a briefly or incompletely monitored case: the event arrived early in the record, so the run-up *is* the time to onset and there is no missing data to recover. An alert already active at a case's first eligible minute is **prevalent**; one that begins later is **incident**. Cases without an event take a sampled pseudo-onset drawn from the event cases' timing distribution, so controls are scored over a comparable window. Every lead time reported here uses this rule unchanged.

Ethics. An institutional review board granted an exemption from review and waived consent for this secondary analysis (Appendix A.1).

3 The reported number tracks the cohort, not only the model

3.1 The number moves with the window, in minutes

The reported lead time moves with the group in every one of the fifteen model-by-cohort cells (Table A2). Splitting detected cases by the length of their available pre-onset window puts the dependence in the metric's own units: median lead rises from the shortest to the longest realized band in **15 of 15** cells, by **+26 to +54 minutes**, each pointwise interval excluding zero, an endpoint contrast of **0.51 to 1.02 minutes of lead per minute of window difference**. A single headline figure therefore describes the cohort's observation-length distribution as much as the model: across its three cohorts the transformer, whose bands are Table A3, reports 5 to 9 minutes of median lead

for its shortest-observed band and 46 to 59 for its longest, and the pooled 17 to 24 minutes sits between them because that is where the median case happens to fall.

Detection is itself a function of window length, so we report that rate rather than conditioning on it silently. Re-cut on all event cases, the detected share moves with window length in every cell, by 0.105 to 0.556, falling in the two families that read the window as a sequence and rising in the three that read a per-minute vector (Table A4). For the transformer that selection runs *against* the reported gradient, which survives it; for the tabular families it runs with the gradient, so part of theirs may be survivorship.

3.2 The same fact as a ratio, in every family

The gradient is universal; its magnitude is not. The median ratio of a case's lead to its own observed window is 0.455 to 0.944 and the Spearman correlation between lead and window length is 0.501 to 0.906 across all fifteen cells, again with no pointwise interval covering zero. The transformer is the limiting case at 0.899 to 0.944 with $\rho = 0.706$ to 0.906, its first alert falling 90 to 94% of the way back to the start of the window it was given; the mildest cells are logistic regression in VitalDB (0.455) and XGBoost in VitalDB (0.467), where the first alert falls nearer the middle of the window than its start. So a claim that the quantity *is* the window fits the transformer and overstates the tabular families.

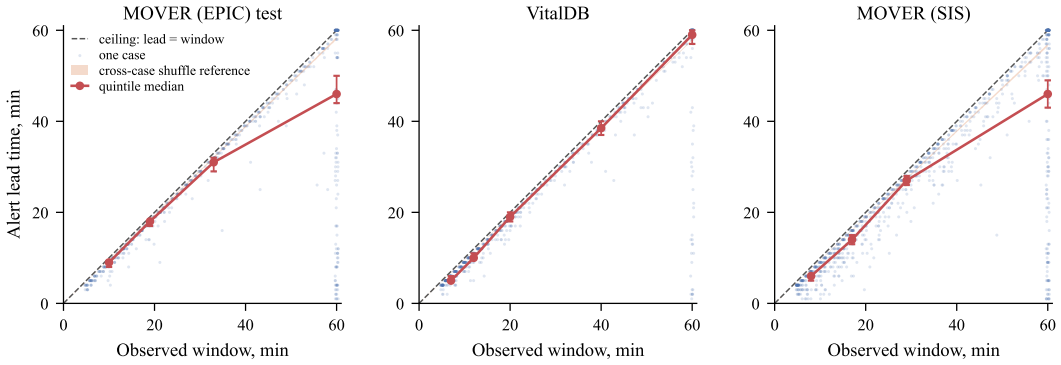


Figure 1: **Alert lead time against the available pre-onset window.** Every detected event case under the ITE Transformer, three held-out cohorts; all fifteen model-by-cohort cells are Table A2, and the five-family scatter is Figure A1. The dashed diagonal is the ceiling. The shaded band is the 2.5th to 97.5th percentile envelope of the cross-case shuffle reference, not a per-case prediction interval. Red markers are the band medians of Table A3 with 95% cluster bootstrap intervals.

3.3 Explicit references for the ceiling dependence

Part of the relationship is built into the definition: a lead time cannot exceed the window it is searched in, so the ratio is bounded in [0, 1] by construction and positive dependence can arise without patient-specific timing. A bound does not specify a distribution, so we use two explicit references rather than treating the diagonal as a null. **Both references are biased *toward* agreement with the observed values, in the direction that favors our own reading, so one that reaches or exceeds the observed value is the expected and uninformative outcome.** The hold variant never falls short; the length-matched variant does, in 6 of 15 cells.

A continuous uniform reference has median 0.5. Placing a pseudo-crossing uniformly on each case's own window gives a reference median of 0.518 to 0.534, which the observed ratio exceeds by **+0.372 to +0.421** for the transformer. It does not exceed it everywhere: in **2 of 15** cells the observed ratio sits *below* the reference and in one more it is indistinguishable from it, all three of which are VitalDB cells (Table A5).

A cross-case shuffle preserves the aggregate relationship (Figure 1). Each recipient keeps its window and reference time but receives another case's whole score sequence, aligned by elapsed window time: generic elapsed-time shape survives, the match between trajectory and recipient does

not. Permutations are derangements, 1,000 per cell. The shuffled median ratio is **0.72 to 0.97** and Spearman **0.63 to 0.94**, so the reference exceeds the observed ratio in **15 of 15** cells, which under the rule above is what an upward-biased reference is expected to do. The length-matched variant is weaker: there the observed correlation is **not distinguishable from the reference in 9 of 15 cells**, and the reference point estimate is the larger within 6 of those 9. In the **remaining 6 cells the observed correlation exceeds the reference and is distinguishable from it** — the three tabular families in the two external cohorts, exactly (Table A5).

Under the hold variant the observed association did not exceed what a window-only reassignment can produce in any cell; under the length-matched variant it did so in 6 of 15. Neither is the same as showing that a window-only process generates it, nor that patient-specific information is absent. Reassignment also changes which event cases are detected, and both variants bias upward, by selecting longer, already-settled donors or holding a short donor's last value.

4 What the reported lead conceals about the operating point

4.1 The transformer clears its threshold almost immediately, in event and control alike

Within three minutes of its first scored minute the transformer's median score is already above the threshold its operating point uses in every cohort, and it is there for event and control cases alike; in MOVER (EPIC) it takes one minute (Figure A2; event-aligned, Figure A3).

In MOVER (EPIC) the median score rises from 0.530 at the first scored minute to 0.938 one minute later in control cases and from 0.541 to 0.951 in event cases, against a transported threshold of **0.894**; by minute five the two groups sit at 0.973 and 0.975. VitalDB clears at minute two and MOVER (SIS) at minute three, in both groups (Appendix A.8).

The consequence is direct: a case whose score settles above threshold in its first monitored minutes registers its first crossing as soon as its alert window opens, so its lead is whatever that window holds. That is the near one-for-one contrast in Table A3.

No other family does this in more than one cohort (Figure A2): the LSTM clears its own threshold in MOVER (EPIC), in both groups, and in neither external cohort; the other three clear it nowhere. Sustained elevation is visible in the share of monitored minutes at or above threshold: **20.0 to 52.3% of the transformer's event minutes and 2.5 to 18.3% of its control minutes, against 0 to 26.7% and 0 to 1.7% for the other four families**. Separately, a cross-family property of the score paths is consistent with this mechanism without establishing its cause (Appendix A.10, Table A6).

The obvious objection is that this model simply alerts continuously, and reporting the operating point beside the lead shows that it largely does (Table A7; composition and controls, Table A8). Its case-level specificity is 0.406, 0.379 and 0.467, so it alerts on 53 to 62% of control cases, and its positive predictive value of 0.625, 0.380 and 0.585 sits 0.065, 0.021 and 0.051 above the prevalence it is measured against. That mechanism is specific to this model and can account for the size of its result. It does not account for the pattern of Section 3, which holds in all fifteen combinations: the claim is that the reported metric is susceptible to the available-window distribution, not that every model reaches that susceptibility by the same score-level route. What this model adds is that **the reported lead carries none of the operating point.** Its median leads, 24, 17 and 22 minutes, are the longest of the five families in two cohorts and tied for longest in the third, and 24 is the longest of the fifteen, while its specificity is the lowest within every cohort and the second-lowest of the fifteen, above only its own 0.379.

That holds beyond one model. **Across the fifteen combinations the reported median lead runs opposite to case-level specificity** (Spearman -0.803; 86 of 99 untied pairs), and opposite to PPV in excess of prevalence at -0.719. We report this descriptively and attach no test to it: the fifteen are not independent replications, since five share each cohort's window distribution and three share each family's fit, and any procedure permuting the pairing across all fifteen would treat as exchangeable units that are not. The ordering is nonetheless present within each cohort taken alone, where that structure is held fixed — 9 of 9 untied pairs in MOVER (EPIC), 8 of 10 in VitalDB and 9 of 9 in MOVER (SIS), Spearman -0.97, -0.70 and -0.97 — though five models per cohort is thin. And a transported threshold does not deliver the same realized sensitivity everywhere: it runs from 0.461 to 0.849 (Table A7), so this is a descriptive association among the operating points as transported, not a sensitivity-matched ranking of models. What it supports is narrower and still worth stating: a

reader ranking these systems by the number they publish would not be ranking them by how usable their operating points are.

4.2 What these findings do and do not license

Both findings are properties of a pipeline as deployed, not of an architecture (Section 7). Nor is the claim that cohort dependence makes the statistic invalid: it is that the median alone suppresses the window distribution a reader needs to interpret it. Neither is a claim about physiology. Mean arterial pressure is one of the model's input channels as well as the quantity the label is computed from, so nothing here shows that a score anticipates a physiological process. Under a pre-specified rule the score does not depart from its own baseline before mean arterial pressure departs from its (Appendix A.13).

5 Two natural responses, and why neither works

A metric with a plausible-looking repair that fails is worse than one with none, because the repair gets adopted.

5.1 Prevalent alerts are not the explanation

The first thought is that the metric is dominated by prevalent alerts. Such cases are **2.6 to 25.5%** of detected event cases across the fifteen cells, the per-cell composition being Figure A4, and recomputing the median lead on incident alerts alone costs **0 to 3 minutes in all fifteen cells**, the transformer included (Appendix A.14). The relationship of Section 3 is therefore carried by the ordinary cases. Prevalence stays low because most detected event cases have a window shorter than 60 minutes, so it opens while the score is still settling: prevalence and a warm-up crossing are the same behavior either side of the first eligible minute.

5.2 Suppressing the first minutes of monitoring is not the fix

The second thought follows from Section 4.1: if the score is unreliable while it settles, discard alerts raised during that period. We implemented exactly that, discarding every minute at or before N minutes of elapsed monitoring and then applying the unchanged threshold.

At $N = 5$, across all five families and three cohorts, the rule costs **3.6 to 6.8 percentage points of case-level sensitivity** and returns **0 to 3.2 percentage points of specificity**. Median lead changes by **−3 to +1 minutes**; in two of the fifteen cells it *increases*, because discarding early minutes removes some short-window cases from the detected set entirely and shifts the median. The trade is poor in every cell, and it does not address Section 3: the surviving alerts still fire at the first opportunity within whatever window remains.

The comparison holds the transported threshold fixed, so it is unmatched on sensitivity by design. Matching is not expected to reverse it: suppression removes alerting minutes and cannot add crossings, so lowering the threshold to restore sensitivity cannot yield more specificity than the fixed-threshold gain above. **We therefore do not recommend the rule**: the problem is not noisy early minutes but the coupling of the reported quantity to the window it was measured in.

5.3 What partially helps, and what remains

Both repairs above change the alert rule and leave the estimand alone. The remaining option reverses that: condition on window length, or treat the first alert as a time-to-event censored when the window ends, as landmarking does [Van Houwelingen, 2007]. Both are computable from the same cached scores, so we report them rather than recommend them (Tables A9 and A10).

A horizon-specific detection rate reduces the dependence rather than removing it. It makes the horizon explicit and conditions the denominator on event cases observed for at least X minutes before onset. That removes one ambiguity of the pooled median, but window length is still not constant: cases observed longer than X contribute different amounts of pre-horizon alerting opportunity. The rate runs 0.278 to 0.692 at a 10-minute horizon and 0.173 to 0.547 at 30 minutes, declining smoothly, a shape a single median suppresses (Table A9). Restricted instead to cases observed for the whole

60-minute window, so the window is a constant, the transformer reports the longest median lead of the five families in all three cohorts, 47, 60 and 57 minutes, while detecting only 0.441, 0.247 and 0.304 of them against 0.762 to 0.910 for XGBoost and the random forest (Table A10). Conditioning removes variation in window length but does not by itself reverse the lead-time ranking; jointly reporting the detected share reveals the corresponding deficit.

Both remain partial. The full-window restriction is the stricter window-constant estimand, but it changes the estimand and the population, discarding 53 to 60% of event cases: the short-window cases a deployed monitor will meet. The horizon-specific curve keeps more cases at short horizons but does not equalize pre-horizon alerting opportunity. Neither repairs the pooled headline median, and the censoring is not independent of the event: short- and long-run-up cases differ in surgery duration and case mix (Appendix A.17).

6 What reporting standards ask, and what they do not

The stakeholders who read a reported lead time, among them a clinician, a regulator, a procurement reviewer and a science writer, meet a number and a name, not the scoring policy behind them, so what a reporting standard asks for limits what any of them can conclude. That is why auditing one metric is a communication result and not only a metrology one.

The authors completed an item-level TRIPOD+AI review for this work, before the analyses of Sections 3 to 5 were conducted, and found no item requesting the disclosures that reveal the dependence. TRIPOD+AI is a general minimum **reporting** standard rather than a validity tool, and one author-completed checklist cannot assess it in either direction. Section 6.2's items are therefore task-specific candidate disclosures that complement the standard rather than correct, invalidate or demonstrate non-adherence to it. Lauritsen et al. [2021] made a request of the same shape — that framing choices be recorded in the TRIPOD statements — and what follows is the continuous-monitoring, first-crossing-timing case of that argument.

6.1 A checklist review that leaves the gap open

Item 12e, performance measures and rationale, is where a lead time is declared: it asks *which* measures were used and *why*, not what else a measure could be responding to. **Item 15 separately addresses model output and threshold selection, while item 23a requests performance estimates with uncertainty.** Table A7 reports lead time beside sensitivity and specificity, a 24-minute lead beside a specificity of 0.406, but neither item explicitly requires a lead-time summary to be jointly presented or interpreted with its realized operating-point metrics. **Items 20a and 20b come closest of all**: both ask for follow-up time among the participant characteristics, and this study answers them with surgery duration and first-onset time. The raw material is therefore requested, as a description of the cohort. What is not requested is the distribution of the window the alert rule searches, nor any pairing of it with the timing statistic it bounds; that window is the variable the reported lead correlates with at $\rho = 0.501$ to 0.906. The remaining near misses are in Table A11.

Item 14, model fairness, is a near miss of a different kind. The checklist's own fairness answer records that arterial-line placement selects toward longer, higher-risk operations. That observation length is not random was therefore **already disclosed**, filed as a limitation, where it reads as a caveat about which patients the findings extend to. Nothing in the standard connects it to the timing metric three tables away.

6.2 Four candidate disclosures for continuously monitored models

Each is tied to a measurement in this paper and computable from artifacts a study of this kind already produces; none requires a new experiment, and none has been validated as a general reporting framework.

1. **Report alert composition by elapsed observation time**: for every operating point, the prevalent, warm-up and settled shares of alerts, as Figure A4 defines them. One stacked bar would have made Section 4.1 visible on sight.
2. **Report a window-conditional timing statistic as the primary timing quantity.** A horizon-specific detection curve may be added as a secondary disclosure, provided its de-

nominator and its remaining dependence on pre-horizon alerting opportunity are stated. Any single median should be secondary and stratified by the available pre-onset windows it was measured over — Section 3.1's spread is what a pooled median hides. Tables A9 and A10 show the two alternatives; Section 5.3 gives their limits.

3. **Report N(t) beside any event-aligned trajectory.** Here the risk set roughly halves between the event and 60 minutes before it (1,404 to 708, 2,804 to 1,433 and 2,641 to 1,374), and the survivors are the long-window ones, the selection of Section 3. **The denominator is not cosmetic**: recomputing the same bins on only the cases observable back to 60 minutes moves the first bin reaching AUROC 0.70 from 15 minutes in all three cohorts to 20, 35 and 20, so the horizon a reader would quote moves by up to twenty minutes on a reporting choice alone, and neither curve is the true one (Figure A5).
4. **Report a path-versus-displacement summary for any score read as a trend**, so that a series traveling much farther than its net displacement is qualified before it is described as rising or falling (Table A6 and Appendix A.10).

The first three are disclosure items in the ordinary TRIPOD sense, naming a quantity that must appear beside one already required; the fourth is closer to a diagnostic, and we offer it as the weakest of the four for that reason.

6.3 What this does and does not generalize to

The specific artifact does not generalize; its shape does: a quantity named for what a clinician would experience, computed from a window whose length varies across patients and is not reported, under a standard that requires the quantity and not the window. Any continuously read model evaluated with a first-crossing statistic over variable available windows is structurally susceptible, and the magnitude has to be measured in each setting. The reporting conclusion is narrow: report a median lead beside the available-window distribution and the operating point that produced it, never alone.

7 Limitations and scope

Retrospective, and one label. Every score is a retrospective forward-running simulation on cached data, not deployment clearance. Both the outcome's level and its duration are conventions we did not vary; the label's dependence on a channel the models read is in Section 4.2.

One model fit, one family of operating points, and one draw of the control reference times. Models were trained at one seed and thresholds come from one validation split at one sensitivity target, transported unchanged; a different fit or target could move Section 5.1's composition, Section 3's magnitudes and Table A7's operating points. Control cases have no onset, so each is assigned a pseudo-onset sampled once at a fixed seed from the event cases' timing distribution. **Every control-side quantity here is conditional on that one draw**: the case-level specificity and positive predictive value of Table A7, the control-side minute share, and the cross-combination association of Section 4.1 that rests on them. Its resampling variability is not propagated into any interval we report: it is one pre-specified draw, and we did not run a sensitivity analysis over it.

Properties of a pipeline, not of an architecture. Section 4.1 describes what one model does under one scoring policy on these cohorts, not what sequence models must do; Section 3's effect is present in all five families, the ratio running 0.455 to 0.944.

The references bias upward. Both shuffle variants make agreement easier to obtain, and they change the detected-case set. They are not estimates of a case-independent process.

A reporting-affordance analysis, and one clinical domain. What we analyze is the affordance of the label as Section 1 defines it, which we take to be the basis on which reporting standards themselves operate. Measured clinician comprehension lies outside that, and is the next study rather than this one; Section 6.2's four items are what it would test. Those four were demonstrated here rather than validated across reporting settings or stakeholder groups, and three cohorts, two health systems and two countries still give one procedure setting and one monitored variable, so they cannot settle whether they are the right four elsewhere.

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Appendix A Methods and complete per-cell results

A.1 Cohorts, models, training and compute

Cohorts and eligibility. Two public datasets supply the three cohorts: VitalDB, and MOVER, whose single release carries both an EPIC-derived and an SIS-derived arm. Both are de-identified. VitalDB v1.0.0 is openly accessible from PhysioNet under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; MOVER is available from the UCI Machine Learning Repository to researchers who sign its data use agreement. Development used MOVER's EPIC arm. The subject-aware split at seed 1004 gave 6,409 training, 1,375 validation and 1,404 test cases out of 9,188, split on unique subject identifiers so that a patient with several operations lands in one fold. The test split holds 1,404 cases, 786 with a sustained event, 1,145 subjects. External evaluation used VitalDB, with 2,804 cases, 1,007 events and 2,730 subjects, and the SIS-derived arm of MOVER, with 2,641 cases and 1,409 events, neither of which contributed to training, calibration or threshold selection. Neither arm of MOVER carries a subject key that resolves against the other, so that statement holds at the level of the released records and cannot be checked at the level of patients. Per-minute eligibility drops the induction prefix, 25 minutes where the record is anchored on anesthesia start, 10 where it is anchored on surgical start, and every minute at or after first onset, so no model is asked to predict an event already in progress. Event cases required five valid pre-onset minutes after that exclusion and controls thirty. Procedure group is never a model input in any family: it separates the cohorts almost perfectly and would be learned as cohort identity.

Training samples, labels and the offset used here. Training uses one sample per case: the 120-minute window ending at that case's reference time, labeled 1 if the case has a sustained event and 0 otherwise. There is no per-minute training target. For an event case the reference time is the first sustained-onset minute shifted back by a lead-time offset, and that offset is **zero for every model reported in this paper**, so the reference time is onset itself. A control has no onset, so it is given a sampled pseudo-onset: its onset ratio, onset divided by case length, is drawn with replacement from the event cases of the same source cohort and multiplied by the control's own case length, then resampled rather than clipped until it falls inside the eligible range, so the control reference times match the event distribution within each source. Event cases with fewer than five eligible pre-onset minutes and controls shorter than thirty are dropped before that sampling. The scored checkpoint is the epoch with the best validation AUROC, and it is the only one used. Evaluation departs from training in the one respect that matters here: every eligible minute of a case is scored independently and inherits its case's label, and `t_to_event` is measured against that same reference time, real onset for an event case and the sampled pseudo-onset for a control.

Features. The same twenty inputs for every family: sixteen per-minute channels, being MAP, systolic and diastolic pressure, their non-invasive counterparts, heart rate, SpO2, end-tidal CO2, temperature, FiO2, MAC, tidal volume, respiratory rate, PEEP and peak inspiratory pressure, and four demographics, being age, sex, height and weight. The families differ in how they consume them. The transformer takes each non-missing cell as a (type, time, value) triplet with demographics as a fixed prefix at time zero, up to 2,048 tokens over a 120-minute window, so missing cells produce no token and are never imputed. The LSTM takes the normalized per-minute window as a value-and-mask pair per channel. Logistic regression, random forest and XGBoost take the last observation carried forward per channel plus the demographics. Per-channel median and interquartile scaling was fitted on the training cases alone and applied unchanged everywhere else.

Architectures and training. Seed 1004 throughout. The transformer is a STraTS-style encoder, a type embedding plus a value MLP plus a time MLP summed per triplet, two encoder layers, model width 128, four heads, feed-forward width 512, dropout 0.1, a learnable classification token and a two-class softmax head; AdamW at learning rate 1.2e-3, weight decay 1e-2 applied to two-dimensional tensors only, gradient clipping at 1.0, a cosine schedule with 5% warmup, batch size 64, cross-entropy, up to 50 epochs with early stopping on validation AUROC at patience 10, under bfloat16 autocast. The LSTM is one layer of width 128 with dropout 0.1, AdamW at 1e-3 with weight decay 1e-2, batch size 64, up to 40 epochs at patience 8. Logistic regression is median imputation and standardization followed by an L2 fit at up to 2,000 iterations. The random forest is 400 trees. XGBoost is 600 rounds at depth 6, learning rate 0.05, row and column subsampling at 0.8, histogram tree method, early stopping after 40 rounds on validation AUC. No hyperparameter search was run over any of these values.

Compute. Every model was trained on one Apple M3 Ultra desktop with 512 GB of unified memory, using PyTorch 2.10 on the Metal backend with bfloat16 autocast; no GPU cluster was used. Wall clock for the five models reported here was 40.3 minutes for the transformer, 14.5 for the LSTM, 3.6 for the random forest, and 34 and 33 seconds for XGBoost and logistic regression. The project trained more than this paper reports: each family was also trained at seven lead-time offsets, of which the transformer arm alone took 3.8 hours. The analyses in this paper add no training and read cached per-minute scores on the same machine. Software was PyTorch 2.10.0, scikit-learn 1.7.1, XGBoost 3.3.0, NumPy 2.2.6, pandas 2.3.2 and SciPy 1.16.1.

Data and code. Both source datasets are public and de-identified. VitalDB v1.0.0 is openly accessible from PhysioNet under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; MOVER is available from the UCI Machine Learning Repository under its data use agreement, one download supplying both arms used here. An institutional review board granted this secondary analysis an exemption from review and a waiver of consent, its identity and protocol number withheld for anonymous review. This work added no data collection and no patient contact. Analysis code is available at an anonymized repository, `https://anonymous.4open.science/r/ioh-leadtime-anon-BD0E`, under the Apache License 2.0. The repository carries the analysis code and the aggregate result files behind the tables and figures, and no patient-level or derived patient-level data of any kind; each dataset is obtained from its own provider under the access and license terms above.

A.2 What the label invites, and what the number supports

Table A1 | What the label invites, and what the number supports. Column one is an affordance of the reported label in the sense Section 1 defines, not measured comprehension. Column two is what this paper's measurement supports in its place, and column three the disclosure that would separate them.

What the label invites	What the number supports here	The disclosure that would close the gap
The model warns about 24 minutes ahead	A median over detected event cases, itself a window-dependent set, under this cohort's window distribution (Section 3.1)	The pre-onset window distribution, and a window-conditional timing statistic (Section 6.2)
Twenty-four minutes is a property of the model	Within one model and cohort, median lead rises across window bands by 26 to 54 minutes, in 15 of 15 cells (Table A2)	Any single median stratified by the realized available window (Section 6.2)
Twenty-four minutes of usable warning	A lead read at case-level specificity 0.406, the lowest of the five families in that cohort (Table A7)	The operating point reported beside the lead (Sections 4.1, 6.3)
A longer lead indicates a better model	A descriptive association running the other way: lead is opposite to specificity across the fifteen combinations, Spearman -0.803 (Section 4.1)	Lead and operating point reported together, not lead alone (Section 6.3)

A.3 Lead time by quantile-derived band of observed window, all fifteen cells

Table A2 | Median alert lead time by quantile-derived band of observed window, all fifteen cells. Main text Section 3.1 reports the ITE Transformer. The gradient is positive in every cell. `Bands` is the number of quantile-derived groups actually realized: where the observed window is censored at the 60-minute alert window the upper quintiles collapse onto a single value, and duplicate edges are dropped rather than bands invented. The gradient is the difference of the highest and lowest band medians, and its interval is the paired cluster bootstrap of that difference: one weight matrix is drawn per cell and reused across bands, and the band edges and the window span are held fixed at their full-sample values so the ratio's denominator is not itself resampled. The interval excludes zero in all fifteen cells. `Lead min per window min` is the ratio of those two between-band median contrasts and not a per-case slope, so a value slightly above 1.0 carries no contradiction with the per-case constraint that a lead cannot exceed its own window.

Model	Cohort	Bands	Window Q1→last, min	Lead Q1→last, min	Gradient, min (95% CI)	Lead min per window min (95% CI)
ITE Transformer	MOVER (EPIC) test	4	10 → 60	9 → 46	+37 [+35, +41]	0.74 [0.70, 0.82]
ITE Transformer	VitalDB	5	7 → 60	5 → 59	+54 [+51, +55]	1.02 [0.96, 1.04]
ITE Transformer	MOVER (SIS)	4	8 → 60	6 → 46	+40 [+37, +43]	0.77 [0.71, 0.83]
LSTM	MOVER (EPIC) test	4	10 → 60	6 → 44	+38 [+35, +41]	0.77 [0.70, 0.82]
LSTM	VitalDB	5	6 → 60	3 → 34	+31 [+21, +48]	0.57 [0.39, 0.89]
LSTM	MOVER (SIS)	4	9 → 60	5 → 31	+26 [+19, +30]	0.51 [0.37, 0.59]
XGBoost	MOVER (EPIC) test	3	12 → 60	8 → 38	+30 [+26, +34]	0.62 [0.54, 0.71]
XGBoost	VitalDB	3	9 → 60	3 → 30	+27 [+22, +30]	0.53 [0.43, 0.58]
XGBoost	MOVER (SIS)	3	12 → 60	5 → 36	+31 [+28, +33]	0.65 [0.58, 0.69]
Random forest	MOVER (EPIC) test	3	13 → 60	8 → 38	+30 [+26, +34]	0.64 [0.55, 0.72]
Random forest	VitalDB	3	9 → 60	3 → 34	+31 [+29, +34]	0.61 [0.57, 0.67]
Random forest	MOVER (SIS)	3	12 → 60	6 → 40	+34 [+31, +36]	0.71 [0.65, 0.75]
Logistic regression	MOVER (EPIC) test	3	11 → 60	8 → 40	+32 [+28, +35]	0.65 [0.57, 0.71]
Logistic regression	VitalDB	3	10 → 60	3 → 28	+26 [+20, +28]	0.51 [0.40, 0.56]
Logistic regression	MOVER (SIS)	3	12 → 60	5 → 38	+33 [+29, +36]	0.69 [0.61, 0.75]

A.4 Lead time by band, the transformer's three cohorts

Table A3 | Median alert lead time inside quantile-derived bands of observed window, ITE Transformer. ITE Transformer, three held-out cohorts, at the transported 80%-sensitivity operating point. Quintile edges are cut within each cell on detected cases; where the observed window is censored at the 60-minute alert window the upper quintiles collapse and the realized band count is reported rather than assumed. `Observed window, min` is the median observed pre-onset window of that band's cases rather than an edge. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). All fifteen cells are Table A2; the detection rate these bands condition on is Table A4. Section 3.1 reports the contrast this table carries.

Cohort	Band	n	Observed window, min	Median lead, min (95% CI)	Lead / window (95% CI)
MOVER (EPIC) test	1	139	10	9 [8, 9]	0.88 [0.85, 0.89]
MOVER (EPIC) test	2	111	19	18 [17, 18]	0.94 [0.94, 0.95]
MOVER (EPIC) test	3	121	33	31 [29, 32]	0.97 [0.96, 0.97]
MOVER (EPIC) test	4	240	60	46 [44, 50]	0.98 [0.96, 0.98]
VitalDB	1	143	7	5 [5, 6]	0.83 [0.83, 0.83]
VitalDB	2	140	12	10 [10, 11]	0.91 [0.90, 0.91]
VitalDB	3	132	20	19 [18, 20]	0.94 [0.94, 0.95]
VitalDB	4	134	40	38 [37, 40]	0.97 [0.97, 0.97]
VitalDB	5	135	60	59 [57, 60]	0.98 [0.98, 1.00]
MOVER (SIS)	1	199	8	6 [5, 6]	0.80 [0.80, 0.82]
MOVER (SIS)	2	179	17	14 [13, 15]	0.86 [0.84, 0.91]
MOVER (SIS)	3	186	29	27 [26, 28]	0.94 [0.93, 0.96]
MOVER (SIS)	4	362	60	46 [43, 49]	0.95 [0.93, 0.97]

A.5 Detection rate by band of observed window, all fifteen cells

Table A4 | Detection rate by band of observed pre-onset window, all fifteen cells. Bands are quantiles of the observed window cut over **all** event cases with a positive observed window, not over the detected subset Table A3 and Table A2 cut on, because a within-band detection rate needs every event case of that window length in its denominator. The two banding schemes are therefore not interchangeable; the band counts here are event-case counts rather than detected counts, and the median leads differ from Table A2's, sometimes by several minutes. The detection rate summed over bands reproduces the case-level sensitivity of Table A7 exactly in all fifteen cells, which is the check that the denominators are the same event cases. Intervals are 95% cluster bootstrap.

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	1	159	159	11	1.000 [1.000, 1.000]	9 [8, 10]
ITE Transformer	MOVER (EPIC) test	2	164	164	23	1.000 [1.000, 1.000]	22 [20, 24]
ITE Transformer	MOVER (EPIC) test	3	150	149	44	0.993 [0.979, 1.000]	42 [40, 43]
ITE Transformer	MOVER (EPIC) test	4	313	139	60	0.444 [0.388, 0.498]	54 [33, 58]
ITE Transformer	VitalDB	1	204	204	8	1.000 [1.000, 1.000]	6 [6, 7]
ITE Transformer	VitalDB	2	203	203	18	1.000 [1.000, 1.000]	16 [14, 17]
ITE Transformer	VitalDB	3	600	277	60	0.462 [0.421, 0.500]	44 [42, 46]
ITE Transformer	MOVER (SIS)	1	287	283	10	0.986 [0.971, 0.997]	7 [7, 8]
ITE Transformer	MOVER (SIS)	2	285	281	25	0.986 [0.971, 0.997]	23 [22, 24]
ITE Transformer	MOVER (SIS)	3	837	362	60	0.432 [0.398, 0.465]	46 [43, 49]
LSTM	MOVER (EPIC) test	1	159	148	11	0.931 [0.889, 0.970]	7 [6, 8]
LSTM	MOVER (EPIC) test	2	164	157	23	0.957 [0.925, 0.987]	20 [18, 21]
LSTM	MOVER (EPIC) test	3	150	141	44	0.940 [0.897, 0.974]	39 [37, 41]
LSTM	MOVER (EPIC) test	4	313	188	60	0.601 [0.545, 0.652]	40 [34, 46]
LSTM	VitalDB	1	204	166	8	0.814 [0.760, 0.868]	3 [3, 4]
LSTM	VitalDB	2	203	147	18	0.724 [0.660, 0.783]	10 [9, 12]
LSTM	VitalDB	3	600	182	60	0.303 [0.268, 0.340]	32 [28, 34]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
LSTM	MOVER (SIS)	1	287	196	10	0.683 [0.628, 0.738]	6 [5, 6]
LSTM	MOVER (SIS)	2	285	188	25	0.660 [0.604, 0.714]	18 [16, 21]
LSTM	MOVER (SIS)	3	837	393	60	0.470 [0.437, 0.504]	33 [29, 36]
XGBoost	MOVER (EPIC) test	1	159	110	11	0.692 [0.619, 0.760]	7 [5, 9]
XGBoost	MOVER (EPIC) test	2	164	132	23	0.805 [0.743, 0.865]	17 [16, 19]
XGBoost	MOVER (EPIC) test	3	150	130	44	0.867 [0.806, 0.918]	32 [28, 36]
XGBoost	MOVER (EPIC) test	4	313	265	60	0.847 [0.806, 0.885]	40 [34, 44]
XGBoost	VitalDB	1	204	110	8	0.539 [0.471, 0.606]	3 [2, 3]
XGBoost	VitalDB	2	203	132	18	0.650 [0.587, 0.718]	7 [5, 9]
XGBoost	VitalDB	3	600	457	60	0.762 [0.728, 0.794]	26 [23, 30]
XGBoost	MOVER (SIS)	1	287	179	10	0.624 [0.570, 0.678]	5 [4, 5]
XGBoost	MOVER (SIS)	2	285	228	25	0.800 [0.753, 0.844]	14 [13, 16]
XGBoost	MOVER (SIS)	3	837	706	60	0.843 [0.819, 0.866]	35 [33, 37]
Random forest	MOVER (EPIC) test	1	159	107	11	0.673 [0.596, 0.746]	6 [5, 8]
Random forest	MOVER (EPIC) test	2	164	135	23	0.823 [0.762, 0.880]	17 [15, 18]
Random forest	MOVER (EPIC) test	3	150	129	44	0.860 [0.800, 0.914]	30 [27, 34]
Random forest	MOVER (EPIC) test	4	313	280	60	0.895 [0.861, 0.928]	41 [35, 45]
Random forest	VitalDB	1	204	140	8	0.686 [0.626, 0.747]	3 [2, 3]
Random forest	VitalDB	2	203	157	18	0.773 [0.711, 0.831]	7 [6, 10]
Random forest	VitalDB	3	600	512	60	0.853 [0.825, 0.882]	32 [30, 35]
Random forest	MOVER (SIS)	1	287	201	10	0.700 [0.646, 0.752]	5 [5, 6]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
Random forest	MOVER (SIS)	2	285	245	25	0.860 [0.819, 0.899]	16 [14, 17]
Random forest	MOVER (SIS)	3	837	750	60	0.896 [0.875, 0.916]	38 [36, 41]
Logistic regression	MOVER (EPIC) test	1	159	119	11	0.748 [0.678, 0.814]	7 [6, 9]
Logistic regression	MOVER (EPIC) test	2	164	138	23	0.841 [0.787, 0.896]	18 [17, 20]
Logistic regression	MOVER (EPIC) test	3	150	134	44	0.893 [0.843, 0.940]	36 [32, 38]
Logistic regression	MOVER (EPIC) test	4	313	267	60	0.853 [0.812, 0.892]	42 [36, 46]
Logistic regression	VitalDB	1	204	71	8	0.348 [0.283, 0.412]	3 [2, 3]
Logistic regression	VitalDB	2	203	95	18	0.468 [0.404, 0.539]	7 [6, 8]
Logistic regression	VitalDB	3	600	298	60	0.497 [0.454, 0.537]	26 [22, 30]
Logistic regression	MOVER (SIS)	1	287	178	10	0.620 [0.566, 0.676]	5 [5, 5]
Logistic regression	MOVER (SIS)	2	285	221	25	0.775 [0.727, 0.821]	14 [12, 16]
Logistic regression	MOVER (SIS)	3	837	714	60	0.853 [0.829, 0.876]	36 [34, 38]

A.6 The relationship in all five families

Figure 1 plots the transformer in three cohorts. The lead-to-window relationship it shows is not particular to that model: in the per-case scatter for MOVER (EPIC) test it is present in every family, with the transformer the limiting case rather than the exception. Figure A1 covers that cohort only; the other two are in Table A2, where the gradient is positive with its interval excluding zero in every cell.

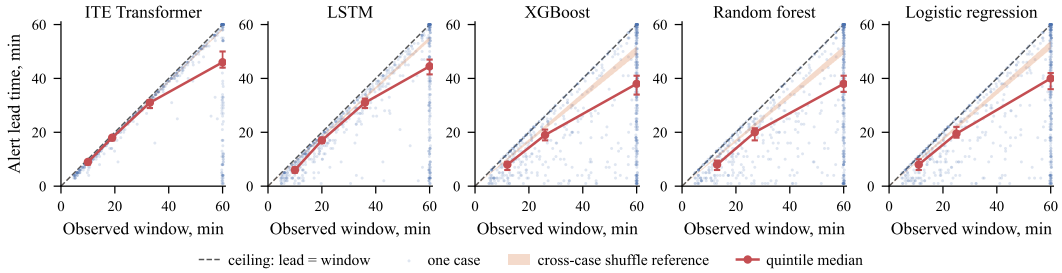


Figure A1: **Alert lead time against observed window length, five model families, MOVER (EPIC) test.** Axes and overlays as in Figure 1. The relationship holds in every family; the transformer is the limiting case rather than the exception.

A.7 Both references, all fifteen cells

Table A5 | Both references, all fifteen cells. Observed ratio against the uniform-placement reference of Section 3.3 and both variants of the cross-case shuffle. The uniform reference's expectation is 0.5 by construction; the two shuffle variants differ only in how a donor shorter than its recipient is handled, and both bias the reference upward. Intervals are the 2.5th and 97.5th percentiles of the reference distribution.

Model	Cohort	Observed ratio	Uniform ref.	Shuffle ref., hold	Shuffle ref., length-matched	Observed ρ	Shuffle ρ, hold	Shuffle ρ, length-matched
ITE Transformer	MOVER (EPIC) test	0.944	0.524 [0.491, 0.567]	0.970 [0.967, 0.974]	0.962 [0.951, 0.971]	0.706	0.888 [0.855, 0.920]	0.783 [0.705, 0.856]
ITE Transformer	VitalDB	0.923	0.532 [0.500, 0.573]	0.966 [0.963, 0.967]	0.957 [0.947, 0.969]	0.906	0.941 [0.926, 0.955]	0.933 [0.890, 0.967]
ITE Transformer	MOVER (SIS)	0.899	0.527 [0.500, 0.567]	0.949 [0.941, 0.950]	0.918 [0.900, 0.938]	0.801	0.877 [0.855, 0.895]	0.775 [0.713, 0.827]
LSTM	MOVER (EPIC) test	0.851	0.522 [0.483, 0.563]	0.913 [0.900, 0.923]	0.866 [0.833, 0.893]	0.650	0.825 [0.784, 0.860]	0.673 [0.596, 0.743]
LSTM	VitalDB	0.619	0.534 [0.500, 0.583]	0.870 [0.850, 0.883]	0.752 [0.667, 0.833]	0.759	0.845 [0.798, 0.886]	0.705 [0.593, 0.804]
LSTM	MOVER (SIS)	0.717	0.524 [0.500, 0.562]	0.873 [0.857, 0.887]	0.700 [0.633, 0.767]	0.501	0.682 [0.636, 0.727]	0.455 [0.368, 0.547]
XGBoost	MOVER (EPIC) test	0.719	0.519 [0.483, 0.558]	0.842 [0.814, 0.867]	0.760 [0.705, 0.800]	0.558	0.701 [0.651, 0.751]	0.581 [0.503, 0.649]
XGBoost	VitalDB	0.467	0.521 [0.483, 0.556]	0.723 [0.667, 0.768]	0.591 [0.517, 0.652]	0.599	0.636 [0.569, 0.698]	0.493 [0.400, 0.580]
XGBoost	MOVER (SIS)	0.600	0.518 [0.500, 0.550]	0.779 [0.750, 0.800]	0.681 [0.633, 0.722]	0.573	0.632 [0.590, 0.673]	0.488 [0.429, 0.541]
Random forest	MOVER (EPIC) test	0.714	0.519 [0.483, 0.556]	0.839 [0.816, 0.867]	0.777 [0.733, 0.817]	0.584	0.714 [0.665, 0.764]	0.599 [0.531, 0.667]
Random forest	VitalDB	0.533	0.523 [0.500, 0.556]	0.764 [0.721, 0.800]	0.685 [0.625, 0.733]	0.651	0.678 [0.622, 0.729]	0.555 [0.479, 0.627]
Random forest	MOVER (SIS)	0.667	0.518 [0.500, 0.550]	0.811 [0.783, 0.833]	0.741 [0.700, 0.783]	0.618	0.659 [0.625, 0.692]	0.536 [0.486, 0.589]
Logistic regression	MOVER (EPIC) test	0.781	0.520 [0.483, 0.557]	0.872 [0.849, 0.898]	0.802 [0.767, 0.842]	0.597	0.739 [0.694, 0.779]	0.612 [0.544, 0.676]
Logistic regression	VitalDB	0.455	0.521 [0.471, 0.567]	0.721 [0.667, 0.781]	0.577 [0.500, 0.650]	0.607	0.642 [0.549, 0.720]	0.486 [0.374, 0.584]
Logistic regression	MOVER (SIS)	0.618	0.518 [0.500, 0.550]	0.782 [0.750, 0.800]	0.734 [0.692, 0.767]	0.597	0.644 [0.607, 0.682]	0.548 [0.495, 0.595]

Three cells in this table are the ones Section 3.3 names as not exceeding the uniform reference, all of them VitalDB. The observed ratio sits below it for XGBoost by 0.054 and for logistic regression by 0.067, and random forest is indistinguishable from it (+0.011, $P = 0.824$). For the transformer the excess over the uniform reference is +0.372 to +0.421. Under the length-matched variant the transformer's ratio reference spans 0.92 to 0.96. Every contrast quoted here is computed from the unrounded values rather than from the rounded columns above, so recomputing one from the table can differ in the third decimal.

A.8 Score against elapsed monitoring time

Section 4.1 reads the first ten monitored minutes of every case, and reports that the ITE Transformer clears its own threshold there in every cohort. The margins for the others are wide. Over those ten minutes the highest median score reached in either group is 0.69 against a threshold of 0.78 for logistic regression in MOVER (EPIC), 0.65 against a threshold of 0.91 for XGBoost, and 0.54 against a threshold of 0.79 for random forest. The LSTM is the one exception: in MOVER (EPIC) it reaches 0.99 in event and 0.99 in control cases against a threshold of 0.97, clearing it in both groups, and it clears it in neither external cohort. This is that view for all fifteen cells, on the elapsed axis rather than the event-aligned one of Figure A3.

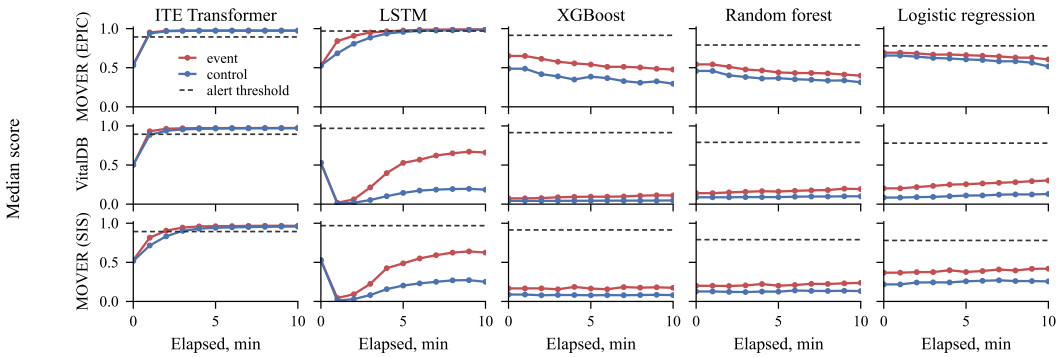


Figure A2: **Median score against elapsed monitoring time, with each family's own alert threshold.** Rows are cohorts, columns are model families. Elapsed time is measured from the case's first scored minute. The dashed rule in each panel is that family's transported raw-score threshold at the 80%-sensitivity target, transferred unchanged from the MOVER (EPIC) validation split. Only the ITE Transformer's early rise clears its own threshold in every cohort, and it does so in event and control cases together; the LSTM does the same in MOVER (EPIC) alone. The elevation ends around minute 20, past the right edge of this axis. This axis is elapsed time; the event-aligned view is Figure A3.

A.9 Event and control score on the event-aligned axis

The shape a reader expects of a continuously read score, rising in the cases that go on to have an event and much flatter in the rest, is **present in all fifteen cells** (Figure A3). Three things about it are worth reporting, and none is visible in a trend statistic.

When the separation arrives runs opposite to near-onset discrimination. For the three tabular families the event and control means are already distinguishable 60 minutes before onset in every cohort, the gap at that range running +0.049 to +0.109 with each pointwise interval excluding zero. For the transformer the gap at 60 minutes **covers zero in all three cohorts** (-0.023 [-0.080, 0.035], -0.014 [-0.057, 0.030] and -0.006 [-0.043, 0.032]), and first excludes it at 35, 20 and 30 minutes. The LSTM sits between the two. The family whose scores separate latest is the one with the highest discrimination near onset.

The event mean rises toward onset, and for the transformer the control mean falls by as much. Over the hour the transformer's event mean rises +0.239, +0.266 and +0.269, which is the tendency a reader expects and it is substantial, while its control mean falls -0.252, -0.263 and -0.196, so about half of the widening is carried by the controls. The LSTM's control mean also falls, by 0.059 to 0.185. For the tabular families the control mean moves between -0.070 and +0.021, rising in several cells, and the widening is almost entirely event-side. So "the score rises in patients who are about to have an event" holds in every cell here; what it leaves out is that for the transformer an equal share of the widening is the control mean coming down, and an event-only trend statistic cannot separate the two.

The split follows the representation more than the architecture. The control side contributes most where the window is read as a sequence and least where the model reads one feature vector per scored minute, carrying each channel's last observation forward, so it sees a level and has little shape to drift with. The ordering is not clean: the tabular families' control means fall in MOVER (EPIC) as well and only flatten in the two external cohorts. Representation and family are not crossed in this design, so this is a reading the data supports rather than a hypothesis it tests. It does say where to look: **the control-side contribution is a thing to measure in models given freer access to the input series**, and reporting the two sides of a widening gap separately is what makes it visible.

Three things this figure is not. The control series is aligned on the sampled pseudo-onset of Section 2 rather than on an event, so its fall is behavior over the last hour of a comparable window and not an approach to anything. Figure A2 plots medians and this figure means, and the two are not interchangeable here. And the two figures use different alignments: Figure A2's axis is elapsed time from the case's first scored minute and covers the first ten minutes of the record, where the transformer has just settled above its own 0.894 threshold, whereas the bins here sit far later in the case, where its event mean is 0.383. **The two are not in conflict, and the length of the plateau is what reconciles them.** The median score holds at or above the transported threshold through elapsed minute 24, 21 and 17 in event cases and 23, 20 and 16 in controls, then falls to 0.03 to 0.31 by minute 30. Figure A2 covers the first ten of those minutes and stops inside the plateau; every bin in this figure falls after it in all but the shortest records.

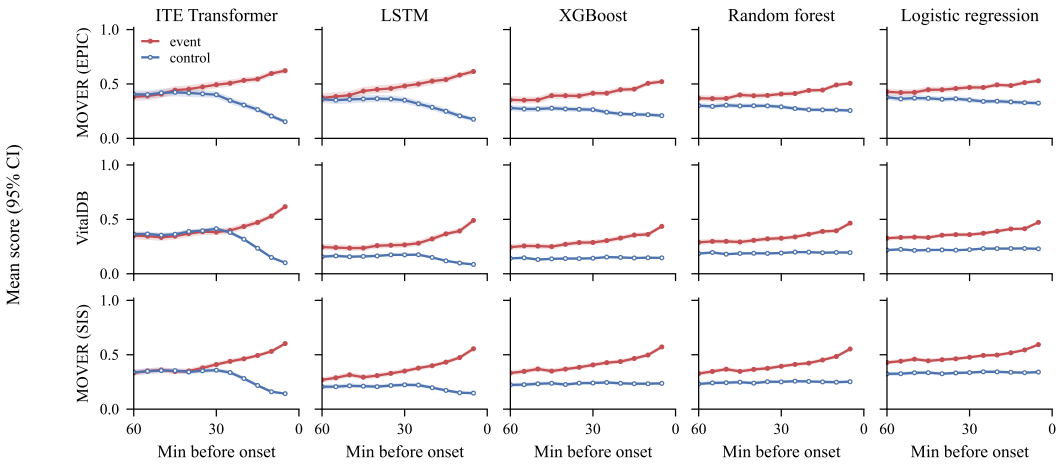


Figure A3: **Mean score in event and control cases against minutes before onset.** Rows are cohorts, columns are model families, the same grid as Figure A2, on the event-aligned axis instead of the elapsed one. Event cases solid and filled, control cases dashed and open; bands are 95% cluster bootstrap. Bins are the 13 centers at five-minute spacing from 60 minutes before onset; the bin at onset is empty by construction, because no minute is strictly before its own reference time, so the series ends at 5 minutes. Control cases are aligned on a sampled pseudo-onset.

A.10 Path decomposition

Table A6 | Path decomposition over the window 60 to 5 minutes before onset. Event episodes have observed scores at both window edges. Directedness is net displacement divided by total distance traveled, so 1 is a monotone climb and 0 is a round trip. Path per net is total distance divided by the absolute net change. Episodes observed at every bin center in the window; median 11 steps per episode. Intervals are 95% cluster bootstrap. Only this window is reported: over 60 to 10 minutes the transformer's net displacement is indistinguishable from zero in two cohorts, and the ratio is then a division by approximately nothing.

Cohort	Model	Episodes	Directedness (95% CI)	Path per unit net
MOVER (EPIC) test	ITE Transformer	313	0.025 [0.002, 0.050]	40.7
MOVER (EPIC) test	LSTM	313	0.043 [0.024, 0.057]	25.9
MOVER (EPIC) test	XGBoost	313	0.049 [0.033, 0.070]	20.0
MOVER (EPIC) test	Random forest	313	0.067 [0.038, 0.093]	16.3
MOVER (EPIC) test	Logistic regression	313	0.061 [0.044, 0.089]	17.6
VitalDB	ITE Transformer	433	0.021 [0.005, 0.038]	146.6
VitalDB	LSTM	433	0.059 [0.035, 0.070]	28.2
VitalDB	XGBoost	433	0.069 [0.049, 0.099]	17.9
VitalDB	Random forest	433	0.075 [0.053, 0.094]	14.4
VitalDB	Logistic regression	433	0.088 [0.063, 0.108]	12.6
MOVER (SIS)	ITE Transformer	670	0.089 [0.066, 0.122]	13.1
MOVER (SIS)	LSTM	670	0.142 [0.118, 0.183]	7.9
MOVER (SIS)	XGBoost	670	0.144 [0.120, 0.167]	7.5
MOVER (SIS)	Random forest	670	0.151 [0.132, 0.168]	7.0
MOVER (SIS)	Logistic regression	670	0.134 [0.119, 0.154]	7.9

Directedness is net displacement divided by total distance traveled: over the window from 60 to 5 minutes before onset it is **+0.021 to +0.151 across all fifteen cells**, each pointwise interval excluding zero, so no family approaches a directed climb. **Directedness describes path shape and does not establish when or whether a particular threshold is touched.** Three further things belong here. The ratio of the median path length to the absolute median net displacement is 7.0 to 146.6 across the fifteen cells; because that ratio becomes unstable as net displacement approaches zero, directedness is the primary summary and both quantities describe path shape rather than prove a crossing mechanism. The transformer sits at the volatile end of both (directedness +0.021 to +0.089, path per unit net displacement 13.1 to 146.6) and logistic regression at the least volatile (+0.061 to +0.134, 7.9 to 17.6). And averaged over cases the score still behaves as a reader expects, rising where an event follows and staying much flatter where none does, but the transformer's event and control means are indistinguishable an hour before onset where the tabular families' are already apart (Figure A3). The property is present under this task and scoring policy and the transformer is its limiting case; the design does not isolate architecture from representation or training.

A.11 Case-level operating point, complete detail

Table A7 | Case-level operating point, all fifteen cells. This is the operating point behind every reported lead time in the paper: every family, every cohort and every interval, and it is where the cross-combination claim of Section 4.1 is checked. Sensitivity and specificity are case-level as defined in Section 2, NNE is alerting cases per detected event and equals 1/PPV, and `Min. above thr.` is the median share of an event case's monitored minutes at or above the transported threshold. Cohort composition, prevalence, PPV in excess of prevalence, the control-side minute share and the share of event cases already alerting by the first elapsed minute are Table A8. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). Section 4.1's cross-combination rank correlations are computed from the unrounded per-cell values rather than from the rounded columns here: recomputing the PPV-lift correlation from PPV and prevalence each rounded to three decimals gives -0.727 instead of -0.719, and the specificity correlation is unchanged at -0.803 either way.

Model	Cohort	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NNE (95% CI)	Min. above thr., events (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	0.777 [0.746, 0.806]	0.406 [0.368, 0.443]	0.625 [0.594, 0.656]	1.60 [1.53, 1.68]	0.523 [0.444, 0.575]	24 [21, 27]
ITE Transformer	VitalDB	0.679 [0.651, 0.707]	0.379 [0.358, 0.403]	0.380 [0.357, 0.402]	2.63 [2.49, 2.80]	0.444 [0.389, 0.500]	17 [15, 19]
ITE Transformer	MOVER (SIS)	0.657 [0.632, 0.682]	0.467 [0.441, 0.494]	0.585 [0.561, 0.609]	1.71 [1.64, 1.78]	0.200 [0.150, 0.238]	22 [20, 24]
LSTM	MOVER (EPIC) test	0.807 [0.779, 0.834]	0.474 [0.433, 0.513]	0.661 [0.631, 0.690]	1.51 [1.45, 1.58]	0.267 [0.209, 0.308]	23 [20, 25]
LSTM	VitalDB	0.492 [0.460, 0.523]	0.812 [0.795, 0.830]	0.595 [0.563, 0.629]	1.68 [1.59, 1.78]	0.000 [0.000, 0.017]	9 [8, 10]
LSTM	MOVER (SIS)	0.551 [0.526, 0.577]	0.709 [0.683, 0.735]	0.685 [0.657, 0.712]	1.46 [1.40, 1.52]	0.017 [0.017, 0.020]	15 [14, 18]
XGBoost	MOVER (EPIC) test	0.810 [0.782, 0.835]	0.608 [0.570, 0.647]	0.725 [0.696, 0.754]	1.38 [1.33, 1.44]	0.117 [0.100, 0.133]	21 [18, 23]
XGBoost	VitalDB	0.694 [0.665, 0.721]	0.754 [0.735, 0.774]	0.613 [0.584, 0.640]	1.63 [1.56, 1.71]	0.067 [0.050, 0.067]	13 [12, 16]
XGBoost	MOVER (SIS)	0.790 [0.768, 0.810]	0.638 [0.610, 0.666]	0.714 [0.691, 0.736]	1.40 [1.36, 1.45]	0.100 [0.100, 0.117]	19 [18, 22]
Random forest	MOVER (EPIC) test	0.828 [0.801, 0.854]	0.555 [0.515, 0.597]	0.703 [0.674, 0.732]	1.42 [1.37, 1.48]	0.150 [0.133, 0.167]	22 [20, 24]
Random forest	VitalDB	0.803 [0.778, 0.827]	0.674 [0.651, 0.695]	0.580 [0.553, 0.604]	1.72 [1.65, 1.81]	0.133 [0.118, 0.150]	15 [12, 18]
Random forest	MOVER (SIS)	0.849 [0.830, 0.867]	0.559 [0.532, 0.586]	0.688 [0.666, 0.710]	1.45 [1.41, 1.50]	0.172 [0.167, 0.183]	22 [20, 25]
Logistic regression	MOVER (EPIC) test	0.837 [0.810, 0.864]	0.518 [0.477, 0.557]	0.688 [0.658, 0.719]	1.45 [1.39, 1.52]	0.139 [0.114, 0.162]	23 [21, 26]
Logistic regression	VitalDB	0.461 [0.431, 0.492]	0.825 [0.808, 0.842]	0.596 [0.565, 0.631]	1.68 [1.59, 1.77]	0.000 [0.000, 0.000]	14 [10, 16]
Logistic regression	MOVER (SIS)	0.790 [0.769, 0.812]	0.585 [0.559, 0.611]	0.685 [0.662, 0.708]	1.46 [1.41, 1.51]	0.117 [0.100, 0.130]	21 [19, 23]

A.12 Cohort composition and the control side

Table A8 | Cohort composition, PPV against prevalence, and the control side. `PPV − prevalence` is the lift over the base rate of the cohort the PPV is measured in, because raw PPV is not comparable across cohorts whose prevalence runs from 0.359 to 0.560. `Alerted by elapsed min 1` is one minus the Kaplan-Meier survival of time to first alert at the first elapsed monitoring minute, censored at the end of each case's own window; it is the censoring-aware form of Section 4.1's settling. It is **not** the prevalent share of Section 5.1, which uses a different numerator and a different denominator; Appendix A.14 sets the two estimands out side by side. Intervals are 95% cluster bootstrap.

Model	Cohort	Event cases	Control cases	Prevalence	PPV − prevalence	Min. above thr., controls (95% CI)	Alerted by elapsed min 1 (95% CI)
ITE Transformer	MOVER (EPIC) test	786	618	0.560	+0.065	0.150 [0.050, 0.233]	0.532 [0.497, 0.565]
ITE Transformer	VitalDB	1007	1797	0.359	+0.021	0.183 [0.163, 0.233]	0.460 [0.428, 0.491]
ITE Transformer	MOVER (SIS)	1409	1232	0.534	+0.051	0.025 [0.017, 0.047]	0.308 [0.285, 0.332]
LSTM	MOVER (EPIC) test	786	618	0.560	+0.101	0.017 [0.000, 0.027]	0.228 [0.198, 0.257]
LSTM	VitalDB	1007	1797	0.359	+0.236	0.000 [0.000, 0.000]	0.087 [0.071, 0.105]
LSTM	MOVER (SIS)	1409	1232	0.534	+0.151	0.000 [0.000, 0.000]	0.138 [0.120, 0.157]
XGBoost	MOVER (EPIC) test	786	618	0.560	+0.165	0.000 [0.000, 0.000]	0.188 [0.161, 0.215]
XGBoost	VitalDB	1007	1797	0.359	+0.253	0.000 [0.000, 0.000]	0.062 [0.047, 0.077]
XGBoost	MOVER (SIS)	1409	1232	0.534	+0.180	0.000 [0.000, 0.000]	0.119 [0.101, 0.136]
Random forest	MOVER (EPIC) test	786	618	0.560	+0.143	0.000 [0.000, 0.000]	0.183 [0.157, 0.212]
Random forest	VitalDB	1007	1797	0.359	+0.221	0.000 [0.000, 0.000]	0.101 [0.082, 0.121]
Random forest	MOVER (SIS)	1409	1232	0.534	+0.154	0.000 [0.000, 0.000]	0.166 [0.147, 0.186]
Logistic regression	MOVER (EPIC) test	786	618	0.560	+0.128	0.000 [0.000, 0.017]	0.226 [0.198, 0.255]
Logistic regression	VitalDB	1007	1797	0.359	+0.237	0.000 [0.000, 0.000]	0.036 [0.025, 0.048]
Logistic regression	MOVER (SIS)	1409	1232	0.534	+0.152	0.000 [0.000, 0.000]	0.121 [0.104, 0.138]

A.13 Departure timing: searched for, not found

We asked whether the score departs from its own baseline before mean arterial pressure departs from its, within an episode, under a pre-specified rule: baseline is the median of the episode's first ten eligible minutes, the robust scale is 1.4826 $\times$ the median absolute deviation over the same minutes, and a departure is the first minute beginning a run of at least three consecutive minutes more than 1.5 scale units in the deteriorating direction.

Score departure does not precede MAP departure. For XGBoost the share of episodes in which the score departs first is significantly *below* one half in all three cohorts. For the transformer the score does not depart at all in most episodes, because its baseline sits at the top of its range: a median baseline of 0.94 to 0.97 leaves median headroom of 0.005 to 0.031, which is less than the bar its own rule sets in all three cohorts, so there is not enough room above baseline for a departure to register. Both facts are reported so the negative cannot be read as an unfair rule. Sensitivity analyses at a 2.0 scale threshold, a five-minute run length, and a search capped at the 60-minute window all leave the direction unchanged.

The question was pre-specified and the answer constrains Section 4: whatever produces the reported lead, it is not the score moving before the monitored variable does.

A.14 Alert composition

Section 5.1 reports the prevalent share as a single number per cell. For the transformer that share is 8.7, 9.6 and 9.0% in the three cohorts, and removing those cases moves its median lead from 24, 17 and 22 minutes to 21, 15 and 19. Splitting detected event cases by when their first crossing fell separates alerts that were already active at the first eligible minute from those raised while the score was still settling and those raised later, which is the disclosure Section 6.2's first item asks for.

That share and Table A8's `Alerted by elapsed min 1` are different quantities rather than two estimates of one. The prevalent share counts detected event cases whose first crossing falls at elapsed minute zero, the first minute of the case's own alert window, over the detected event cases of that cell: 53 of 611, 66 of 684 and 83 of 926 for the transformer. Table A8's column counts event cases whose first crossing falls at elapsed minute zero **or** minute one, over every event case with a positive observed window, detected or not, as one minus the Kaplan-Meier survival of time to first alert with each case censored at the end of its own window: 418 of 786, 463 of 1,007 and 434 of 1,409. Numerator, denominator and conditioning set all differ, and the extra minute is the larger of the differences: minute one alone adds 365, 397 and 351 cases, which is Section 4.1's settling read on the elapsed axis and not a second estimate of the same immediate-alert probability. No case is censored before elapsed minute one in any of the three cohorts, so the survival estimate at that mark equals the crude share.

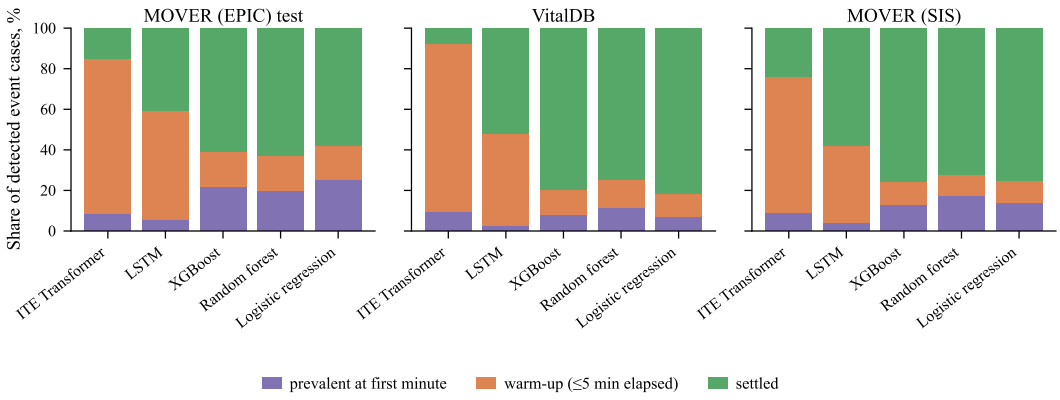


Figure A4: **Composition of detected event cases by elapsed monitoring time at the first crossing.** Prevalent cases are above threshold at their first eligible minute; warm-up cases cross within five minutes of it; settled cases cross later. Five model families, three cohorts, at the transported 80%-sensitivity operating point. The composition differs sharply across families, in that the transformer's alerts are concentrated in the first five minutes, the LSTM's are partly so and the tabular families' are not, while the lead-to-window relationship of Section 3 holds in all of them. That is why Section 4 separates the transformer's direct mechanism from the cross-family directedness result.

A.15 Replacement estimands, all horizons

Table A9 | Fixed-horizon detection rate at all seven horizons, and time to first alert. The fixed-horizon detection rate is the share of event cases observed for at least the stated number of minutes before onset whose first alert came at least that many minutes before onset. Point estimates are shown here; each is a share of the cases in its own column, and its 95% cluster bootstrap interval is computed exactly as elsewhere in this appendix. Section 5.3 quotes the range across cells at the 10- and 30-minute horizons. `n observed` is the denominator at the first and last horizon, and its fall is the reason a single median is not comparable across cohorts. The 60-minute column is the share of fully observed cases that alerted at the very first opportunity.

Model	Cohort	5 min	10 min	15 min	20 min	30 min	45 min	60 min	n observed $\geq 5 \rightarrow \geq 60$
ITE Transformer	MOVER (EPIC) test	0.733	0.690	0.646	0.605	0.512	0.339	0.170	786 → 311
ITE Transformer	VitalDB	0.638	0.560	0.509	0.463	0.390	0.261	0.154	1007 → 429
ITE Transformer	MOVER (SIS)	0.594	0.547	0.504	0.469	0.377	0.247	0.125	1409 → 665
LSTM	MOVER (EPIC) test	0.721	0.663	0.635	0.597	0.490	0.331	0.109	786 → 311
LSTM	VitalDB	0.343	0.278	0.265	0.223	0.173	0.102	0.030	1007 → 429
LSTM	MOVER (SIS)	0.453	0.392	0.346	0.319	0.253	0.167	0.047	1409 → 665
XGBoost	MOVER (EPIC) test	0.688	0.651	0.615	0.575	0.484	0.367	0.119	786 → 311
XGBoost	VitalDB	0.502	0.470	0.453	0.425	0.366	0.237	0.054	1007 → 429
XGBoost	MOVER (SIS)	0.656	0.597	0.565	0.520	0.460	0.326	0.080	1409 → 665
Random forest	MOVER (EPIC) test	0.709	0.674	0.638	0.611	0.506	0.393	0.138	786 → 311
Random forest	VitalDB	0.589	0.559	0.559	0.530	0.478	0.347	0.103	1007 → 429
Random forest	MOVER (SIS)	0.736	0.674	0.640	0.600	0.526	0.403	0.114	1409 → 665
Logistic regression	MOVER (EPIC) test	0.726	0.692	0.658	0.630	0.547	0.401	0.122	786 → 311
Logistic regression	VitalDB	0.338	0.307	0.303	0.274	0.233	0.137	0.026	1007 → 429
Logistic regression	MOVER (SIS)	0.664	0.611	0.573	0.546	0.474	0.353	0.093	1409 → 665

A.16 The full alert window as a constant

Table A10 | Event cases observed for the whole 60-minute alert window. Restricting to these cases makes the observation window a constant rather than a covariate, which is the full-window restriction of Section 5.3. `n` is the number of event cases observed for the whole window and `Detected` the share of them that alert at all. The censoring is not independent of the event, because for an event case a short observable run-up is an early-onset event rather than missing data, so this describes what the rule did under the observation these cohorts supplied. Intervals are 95% cluster bootstrap. For the transformer in MOVER (EPIC) the median lead here differs from Table A4's top band by seven minutes although the two populations differ by two cases, because the detected full-window leads leave a gap between 47 and 54 minutes: 53 of 137 sit at the 60-minute ceiling and the lower tail is sparse, so a one-rank shift in the median crosses the gap.

Model	Cohort	n	Detected	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	311	0.441	47 [33, 59]
ITE Transformer	VitalDB	429	0.247	60 [60, 60]
ITE Transformer	MOVER (SIS)	665	0.304	57 [51, 59]
LSTM	MOVER (EPIC) test	311	0.598	40 [34, 46]
LSTM	VitalDB	429	0.200	33 [21, 45]
LSTM	MOVER (SIS)	665	0.433	28 [22, 33]
XGBoost	MOVER (EPIC) test	311	0.846	40 [34, 44]
XGBoost	VitalDB	429	0.762	33 [30, 36]
XGBoost	MOVER (SIS)	665	0.854	37 [34, 41]
Random forest	MOVER (EPIC) test	311	0.894	41 [35, 45]
Random forest	VitalDB	429	0.851	40 [36, 44]
Random forest	MOVER (SIS)	665	0.910	42 [39, 45]
Logistic regression	MOVER (EPIC) test	311	0.852	42 [36, 45]
Logistic regression	VitalDB	429	0.490	31 [26, 37]
Logistic regression	MOVER (SIS)	665	0.857	40 [37, 43]

A.17 Risk-set composition, and what fixing the cohort does

The population trajectory scores every case that has a minute in the bin, so its denominator shrinks and its composition shifts as the bin moves away from onset. For an event case the observable run-up *is* its time to onset, so a short run-up is not missing data. It is an early-onset event, and the far bins keep only the long run-ups. That is the same selection that drives Section 3.

Fixing the cohort moves the trajectory in both directions (Figure A5). Recomputing the same bins on the cases observable back to 60 minutes raises the far end and lowers the near end, and the two curves cross at 16.4, 12.6 and 14.9 minutes before onset. The first bin reaching AUROC 0.70 moves from 15 minutes in all three cohorts to 20, 35 and 20. Neither curve is the true one: which denominator is used is a reporting choice, it changes the horizon a reader would quote by up to twenty minutes, and it is not currently disclosed. That is the whole of Section 6.2's third item.

Short and long run-ups are different operations, which is why Section 5.3 calls this censoring non-independent. Splitting event cases at a 40-minute observable run-up, the long group's operations are longer in every cohort, by a standardized mean difference of 0.394, 0.515 and 0.428, the largest imbalance on the list in all three; procedure group and ASA class are also imbalanced beyond 0.1 in each. So a case with a short run-up is not a briefly watched version of a case with a long one.

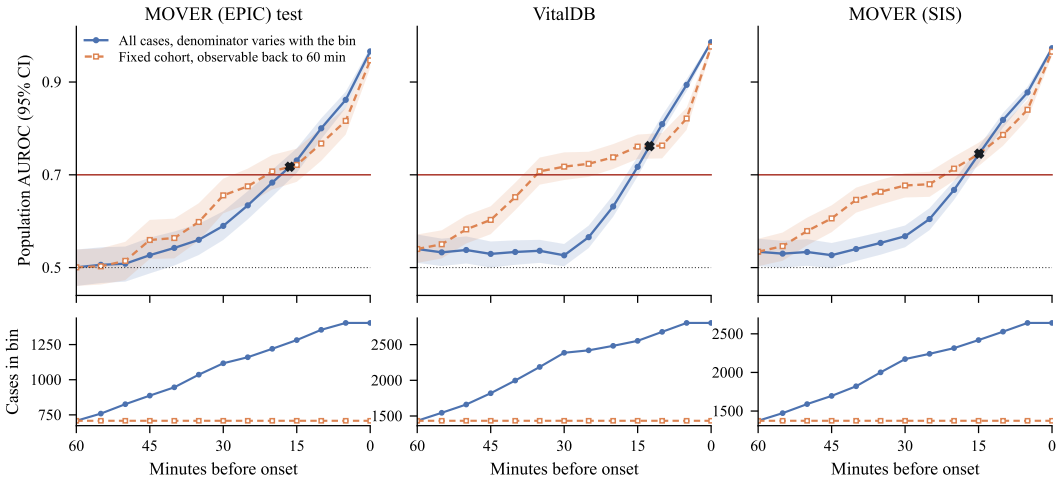


Figure A5: **Population AUROC with its own risk set, event-aligned.** ITE Transformer, 2-minute bins from 60 minutes before onset to onset, in three held-out cohorts. Solid and filled is every case with a minute in the bin, the convention in the published literature, whose denominator varies; dashed and open is the cohort fixed to cases observable back to 60 minutes. Bands are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). The lower row is the number of cases contributing to each bin, on its own scale per cohort. The dotted line marks 0.5, the solid rule 0.70, and the cross marks where the two curves meet.

A.18 The reporting items that came close

Table A11 | The TRIPOD+AI items that came close and missed. Main text Section 6.1 discusses items 12e, 15, 23a, 20a, 20b and 14; this table records the remaining items, together with the detail behind those the main text discusses. The item numbering is that of the TRIPOD+AI statement [Collins et al., 2024]. This is an author-completed self-assessment of one study, not a validity assessment of the standard.

Item	What it asked for	What it did not ask for
12f and 12g, recalibration and how evaluation predictions were calculated	Any recalibration arising from evaluation, and how evaluation predictions were computed: calibration and thresholds fitted on the development validation split and transported unchanged, which is reported	Anything about what that disciplined choice implies; it is the exact setting in which Section 3 holds, so doing the right thing here does not help
20a and 20b, participant flow and characteristics	Participant flow and characteristics *including follow-up time*, answered here with surgery duration and first-onset time	The distribution of the window the alert rule actually searches — the run-up after the induction exclusion, capped at the 60-minute alert window — or any pairing of it with the timing statistic it bounds. This is the closest the standard comes: it asks for the raw material as a cohort descriptor, and never connects it to the timing metric it constrains
21, event counts	The number of participants and outcome events *in each analysis*, which are reported	The same counts broken out by band of available window. The item's "in each analysis" clause reaches development, tuning and evaluation, not the window strata of a timing statistic
8c, blinding of outcome assessment	A blinding statement; answered "not applicable" because the outcome is computed from a monitored channel that is also a predictor	The closest the checklist comes to Section 4, and still about blinding rather than about what a timing metric measures
15, model output	Details and rationale for classification, and how any threshold was identified; answered with the transported threshold itself	Joint presentation or interpretation of a lead-time summary with the realized operating-point metrics that threshold defines, which is where the artifact lives (Table A7)

A.19 Multiplicity

The two references of Section 3.3 form one Benjamini–Hochberg family of 75 formal tests, adjusted within that family, of which 57 are significant at $q < 0.05$. The stratification of Section 3.1 produces intervals rather than tests and does not enter the family. Every bootstrap interval in this paper, including the fifteen window-gradient intervals of Table A2, is a **pointwise descriptive interval**: no family-wise coverage claim is made across the fifteen contrasts, and "each interval excludes zero" is fifteen separate statements rather than one simultaneous one.

A.20 Seeds and reproducibility

Cluster bootstraps use 2,000 replicates, with the clustering unit the subject for two cohorts and the case for MOVER (SIS), whose released subject key is a copy of the case identifier. The two references of Section 3.3 draw 1,000 permutations per cell. The cross-combination rank correlations of Section 4.1 are reported as **point estimates with no accompanying test**. A permutation of the pairing was computed while this analysis was being developed, and is not reported: its unit is a cell, and cells sharing a cohort or a model family are not exchangeable, so the resulting P value would overstate what fifteen dependent combinations can establish. Section 4.1 gives the untied-pair counts and the within-cohort correlations in its place. Model training and the reference-time construction are fixed at a single seed, reported in Appendix A.1.

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